National University of Computer & Emerging Sciences, Karachi
Spring-2022, FAST School of Computing
Re-Midterm-1 Examinations
March 29, 2022, 10:00 am – 11:00 am

Course Code: CS3001	Course Name: Computer Networks
Instructor Name: Dr. Farrukh Salim , Dr. Aqsa Aslam, Mr. Shoaib Raza and Ms. Eman Shahid	
Student Roll No:	Section:

Instructions:

- Return the question paper. Do not write anything on question paper, except your Roll # & Section #.
- Read each question completely before answering it. There are **4 questions and 2 pages.**
- In case of any ambiguity, you may make assumptions. However, your assumptions should not contradict any statement in the question paper.
- All the answers must be solved according to the sequence given in the question paper.
- This paper is subjective. Write the answers only on answer sheet.

Time: 60 minutes. **Max Points**: 50

Question 1: [5 points]

In a given network as Figure 1, two hosts are communicating with each other. Each host is connected using the WiFi 802.11n (450Mbps) with the wireless access point. The wireless access points in turn are connected to the same single router using Fast Ethernet (100Mbps) connection. Assuming there is no other data communicating at the same time. Which one is (are) the bottlneck link(s) between Host-1 and Hsot-2 and why? What is (are) the data rate (in Mbps) of the bottleneck link(s)?

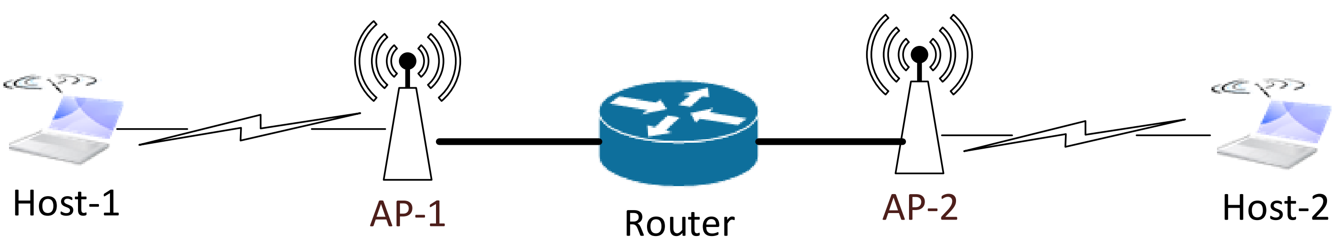

Figure 1

Solution:

WiFi 802.11n operates at 450Mbps, and Fast Ethernet operates at 100Mbps. Thus the bottleneck links are the fast ethernet connections between Access Points and router.

Question 2: [10 points]

Suppose that each Wireless Access Point is at a distance of 200km from the Router (as shown in Figure 1). Further suppose that there is no node processing delay, nor queuing delay at any device. Moreover, also assume that there is no propagation delay between each host and its Access Point. The data rate is assumed to be 2Mbps between each host and Access Point and 10Mbps between each Access Point and the Router. Find the approximate end-to-end delay between Host-1 and Host-2 for a data packet of size 2Mbits.

Solution:

Since it is known that:

d_{prop}: propagation delay:

- d: length of physical link
- s: propagation speed (~2×10^8 m/sec)
- $d_{prop} = d/s$

d_{trans}: transmission delay:

- L: packet length (bits)
- R: link *transmission rate (bps)*
- $d_{trans} = L/R$

The delay encountered are:

- Transmission delay for 2 Mbps link: L / R = 2Mb / 2 Mbps = 1 sec
- Transmission delay for 10 Mbps link: L / R = 2Mb / 10 Mbps = 0.2 sec
- Propagation delay for 10 Mbps link: d / s = 200km / 2×10^8 m/sec = 0.001 sec

As there are two links with 2Mpbs and two links with 10Mbps data rate, thus, total end-to-end delay is: 1.201 * 2 = 2.402 seconds.

Question 3: [5 * 3 = 15 points]

Make a table, that provides transport service requirements in terms of (i) data loss, (ii) throughput and (iii) time sensitiveness of minimum five (5) different network applications.

Solution:

Application	Data Loss	Throughput	Time-Sensitive
File transfer/download	No loss	Elastic	No
E-mail	No loss	Elastic	No
Web documents	No loss	Elastic (few kbps)	No
Internet telephony/ Video conferencing	Loss-tolerant	Audio: few kbps–1 Mbps Video: 10 kbps–5 Mbps	Yes: 100s of msec
Streaming stored audio/video	Loss-tolerant	Same as above	Yes: few seconds
Interactive games	Loss-tolerant	Few kbps–10 kbps	Yes: 100s of msec
Smartphone messaging	No loss	Elastic	Yes and no

Figure 2.4 ♦ Requirements of selected network applications

Question 4: [10 * 2 = 20 points]

For the result of the traceroute command shown below in Figure 2, give answers for below questions:

A. If traceroute command runs again, will the IP address shown by point A will be always this same for google.com? If no, and any other IP address is also possible, then why?
B. What this '30 hops' represents?
C. Which device is represented by the IP address of Point C in the network?
D. What * represents?
E. On running traceroute command, the result for this line shown by point E is always 'Request timed out'. What could be the possible reason behind this?
F. What these three different values represents? and why not only one value suffices?
G. Why these values are mostly smaller than most of the values above? Give reason.
H. What is the importance of this number 13 shown by point H?
I. What this URL show by point H represents?
J. Which type of Internet Control Message Protocol (ICMP) messages are used by traceroute ping command.

```
C:\Users\Public>tracert google.com

Tracing route to google.com [172.217.19.174]   A
over a maximum of 30 hops:
                  B
  1     1 ms     1 ms     1 ms  192.168.43.1   C
  2 D   *        *        *     Request timed out.  E
  3    35 ms    49 ms    44 ms  10.81.77.165
  4    51 ms    36 ms    49 ms  10.81.73.137
  5   179 ms    50 ms    47 ms  10.180.76.1
  6    40 ms    75 ms    70 ms  10.81.171.21
  7 F 522 ms    67 ms   138 ms  116.71.129.225
  8    60 ms    36 ms    40 ms  10.253.4.44
  9 G  58 ms    78 ms    65 ms  10.253.4.8
 10    89 ms    86 ms    90 ms  74.125.118.170
 11    79 ms    55 ms    58 ms  172.253.51.205
 12    57 ms    56 ms    61 ms  209.85.250.17
H 13    56 ms    57 ms    55 ms  zrh04s07-in-f174.1e100.net [172.217.19.174]

Trace complete.                         I
```

Figure 2

Solution:

A. No, because google.com runs on many servers, so any one IP address will be used from the pool of the IP address, which will not necessarily be the same as previous IP address. Moreover, anycast routing is used, in which any one google server will reply to the request.
B. This represents that if the number of hops with destination node is more than 30 number of hops, then this traceroute command or the connection will not run.
C. The first router or the gateway address.
D. * means the reply has not received within timeout.
E. Because the second device has restricted to reply to ping commands.
F. These represent the Round Trip Time (RTT). Three values are taken to measure average behavior.
G. Every router RTT varies mostly due to the variable queuing delays.
H. This represent the last hop. Because it is the maximum number of total hops encountered with the server.
I. This is the real URL name of the server, called canonical name.
J. Echo Request and Echo Reply.